Low-Light Video Enhancement via Negative Image Dehazing
透過負影像去霧技術改善低光影片品質

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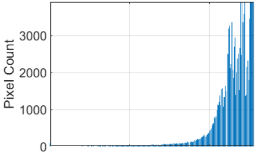
Motivation

In daily life, we often encounter underexposed or low-light images. This led to the hypothesis that by applying an inverse transformation (negative) to a Low-Light image, it would statistically resemble a hazy image. Consequently, we can leverage established image dehazing techniques to restore visibility and details.

Histogram

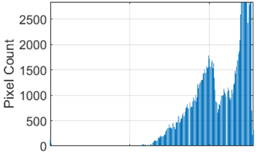
Low-Light Image

Negative Image



Negative Image of hist

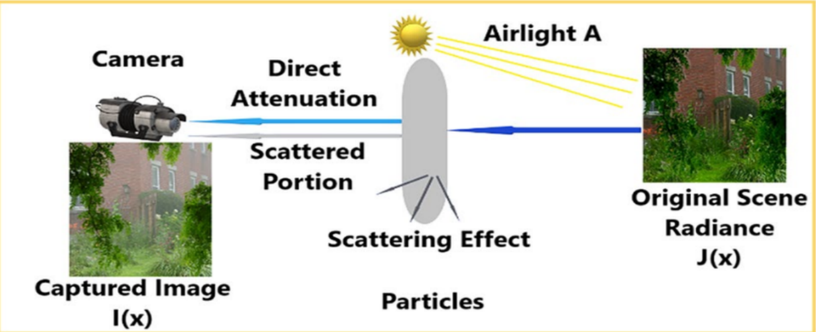
Hazed Image



Hazed Image pf hist

ASM

The **Atmospheric Scattering Model** is a physical model that describes how light attenuates and scatters in environments filled with suspended particles, such as haze or fog.



ASM

The formation of a hazy image is typically described by the Atmospheric Scattering Model:

$$\mathcal{I}(x) = \mathcal{J}(x)t(x) + \mathcal{A}(1 - t(x))$$

where each term represents a specific physical component:

- $\mathcal{I}(x)$ **(Observed Intensity):** The hazy image. In low-light enhancement, this is the negative of the original image.
- $\mathcal{J}(x)$ **(Scene Radiance):** The clear, haze-free image we aim to recover as the final output.
- $t(x)$ **(Transmission Map):** Defined as $e^{-\beta d(x)}$, representing the portion of light reaching the camera through the medium; $0 \leq t(x) \leq 1$
- $\mathcal{A}$ **(Atmospheric Light):** The global ambient light intensity representing the source of scattered light.

Goal: To recover $\mathcal{J}(x)$ by accurately estimating $\mathcal{A}$ and $t(x)$ from the input $\mathcal{I}(x)$.

DCP

The Dark Channel Prior is based on the observation that in most non-haze outdoor images, at least one color channel has some pixels whose intensity is very low and close to zero:

$$\mathcal{J}_{dark}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{R,G,B\}} \mathcal{J}^c(y) \right) \approx 0$$

Why is it close to zero?

- **Shadows:** Natural shadows in images (e.g., crevices, occlusion).
- **Colorful Objects:** Vivid colors (e.g., a red apple) have very low values in other channels (blue/green).
- **Dark Objects:** Naturally dark surfaces like tree trunks or dark soil.

DCP for t(x)

To derive the **transmission map** $t(x)$, we consider the atmospheric scattering model for each color channel $c \in \{R, G, B\}$, assuming that the transmission is constant within a local patch $\Omega(x)$, denoted as $\tilde{t}(x)$:

$$\frac{I^c(y)}{A^c} = \tilde{t}(x)\frac{J^c(y)}{A^c} + 1 - \tilde{t}(x), \quad \forall y \in \Omega(x)$$

Next, we apply the minimum operator across the color channels and the local patch sequentially:

$$\min_{y\in\Omega(x)}\left(\min_c \frac{I^c(y)}{A^c}\right) = \tilde{t}(x)\left(\min_{y\in\Omega(x)}\left(\min_c \frac{J^c(y)}{A^c}\right)\right) + 1 - \tilde{t}(x)$$

DCP for t(x)

According to the Dark Channel Prior, the **dark channel** of a clear haze-free image J approaches zero ($J^{dark}(x) \approx 0$). Consequently, the first term on the right side vanishes:

$$\min_{y \in \Omega(x)} \left(\min_{c} \frac{I^c(y)}{A^c} \right) \approx 1 - \tilde{t}(x)$$

By rearranging the equation and introducing a constant dehazing factor ω $(0 < \omega < 1)$ to retain a small amount of haze for a more natural depth appearance, we obtain the initial estimation of $t(x)$ as:

$$\tilde{t}(x) = 1 - \omega \min_{y \in \Omega(x)} \left(\min_{c \in \{R,G,B\}} \frac{I^c(y)}{A^c} \right)$$

In this project,I choose the $\omega = 0.95$.

DCP for A(x)

The **atmospheric light** $\mathcal{A}$ represents the intensity of the ambient light. Instead of picking the brightest pixel in the original image, we use the **Dark Channel** to avoid interference from white objects.

Estimation Steps:

1. Pick the top 0.1% brightest pixels in the dark channel image $\mathcal{J}_{dark}$.
2. These pixels represent the most haze-opaque regions.
3. Identify the corresponding pixels in the input image $\mathcal{I}$.
4. Select the average (or highest) intensity among these pixels as $\mathcal{A}$.

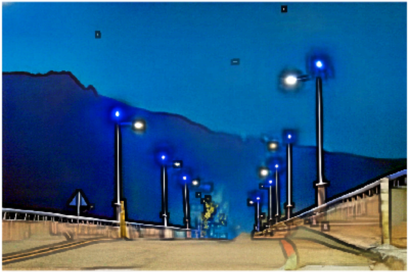
In this project,I choose the average intensity.

Guided Filter

The **Guided Filter** is an edge-preserving smoothing filter. It uses a "Guidance Image" to determine the filtering output, allowing the output image to inherit the structural and edge information of the guidance image.

Why use Guided Filter?

The initial transmission map $t(x)$ is calculated patch by patch, which often leads to blocky artifacts near sharp edges.



Guided Filter

Mathematical Formulation: We assume the refined transmission q is a linear transformation of the guidance image I (the original negative image) within a local window ω_k:

$$q_i = a_k I_i + b_k, \quad \forall i \in \omega_k$$

Where:

- a_k, b_k are the linear coefficients assumed to be constant in ω_k.
- This model ensures $\nabla q = a \nabla I$, meaning q captures the exact edge structures of I.

Guided Filter

To find the optimal coefficients (a_k, b_k), we minimize the following cost function with regularization:

$$E(a_k, b_k) = \sum_{i \in \omega_k} \left((a_k I_i + b_k - t_i)^2 + \epsilon a_k^2 \right)$$

Mathematical Precision:

- **Data Fidelity:** The term $(a_k I_i + b_k - t_i)^2$ ensures the refined output q remains close to the initial estimate t.
- **Regularization:** The parameter ϵ (Degree of Smoothing) penalizes large a_k to prevent overfitting and control smoothness.
- **Structure-Preserving:** The optimization results in a refined transmission map that eliminates blocky artifacts while precisely **preserving object boundaries**.

Guided Filter

To find the optimal solution, we take the partial derivatives of E with respect to b_k and a_k, and set them to zero. First, differentiating with respect to b_k yields:

$$\frac{\partial E}{\partial b_k} = \sum_{i \in \omega_k} 2(a_k I_i + b_k - p_i) = 0$$

This can be simplified by expanding the summation over the window:

$$a_k \sum_{i \in \omega_k} I_i + |\omega| b_k - \sum_{i \in \omega_k} p_i = 0$$

Dividing the entire equation by the number of pixels $|\omega|$ in the window, we express b_k as:

$$b_k = \bar{p}_k - a_k \mu_k$$

Guided Filter

where μ_k and $\bar{p}_k$ represent the mean values of I and p within ω_k, respectively. Next, taking the partial derivative with respect to a_k gives:

$$\frac{\partial E}{\partial a_k} = \sum_{i \in \omega_k} 2(a_k I_i + b_k - p_i) I_i + 2\epsilon a_k = 0$$

Substituting the expression for b_k into this equation yields:

$$\sum_{i \in \omega_k} (a_k I_i + \bar{p}_k - a_k \mu_k - p_i) I_i + \epsilon a_k = 0$$

By expanding and grouping the terms with a_k, we obtain:

$$a_k \left(\sum_{i \in \omega_k} I_i^2 - \mu_k \sum_{i \in \omega_k} I_i \right) + \epsilon a_k = \sum_{i \in \omega_k} I_i p_i - \bar{p}_k \sum_{i \in \omega_k} I_i$$

Guided Filter

By dividing the equation by $|\omega|$ and utilizing the definitions of variance σ_k^2 and covariance σ_{Ip}, the closed-form solution for the coefficients within a specific window ω_k is given by:

$$a_k = \frac{\frac{1}{|\omega|}\sum_{i\in\omega_k} I_i p_i - \mu_k \bar{p}_k}{\sigma_k^2 + \epsilon}, \quad b_k = \bar{p}_k - a_k \mu_k$$

Since a single pixel i is covered by multiple overlapping windows, it receives multiple sets of (a_k, b_k). To ensure a smooth result and eliminate artifacts, we average these coefficients to obtain $\bar{a}_i$ and $\bar{b}_i$. The final output is:

$$q_i = \bar{a}_i I_i + \bar{b}_i$$

J(x)

After estimating the global atmospheric light A and refining the transmission map

$$t(x) = \tilde{t}(x) \to q$$

via the Guided Filter, we can finally recover the clear scene radiance $J(x)$ by inverting the Atmospheric Scattering Model:

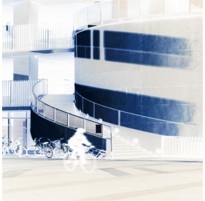
$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A$$

Why use max and What is t_o?

let t_0 = 0.1 beacause we affraid the t(x) close to o and make this division to big

Result

Low-Light Image



Negative Image

Dark Channe

Resulting Image

+Guided Filter

[Comparison of Transmission Maps]

Reference

1. X. Dong, et al., " Fast efficient algorithm for enhancement of low lighting video ", *2011 IEEE International Conference on Multimedia and Expo, Barcelona,* 2011, pp. 1-6
2. K. He, J. Sun, and X. Tang, " Single image haze removal using dark channel prior ", *IEEE Transactions on Pattern Analysis and Machine Intelligence,* 33 (2011), pp. 2341-2353
3. S.-Y. Yang, " Single image dehazing, Lecture Slides ", *Lecture Slides,* 2025